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Bad News? Send an AI. Good News? Send a Human

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Abstract

The present research demonstrates how consumer responses to negative and positive offers are influenced by whether the administering marketing agent is an artificial intelligence (AI) or a human. In the case of a product or service offer that is worse than expected, consumers respond better when dealing with an AI agent in the form of increased purchase likelihood and satisfaction. In contrast, for an offer that is better than expected, consumers respond more positively to a human agent. The authors demonstrate that AI agents, compared with human agents, are perceived to have weaker intentions when administering offers, which accounts for this effect. That is, consumers infer that AI agents lack selfish intentions in the case of an offer that favors the agent and lack benevolent intentions in the case of an offer that favors the customer, thereby dampening the extremity of consumer responses. Moreover, the authors demonstrate a moderating effect, such that marketers may anthropomorphize AI agents to strengthen perceived intentions, providing an avenue to receive due credit from consumers when the agent provides a better offer and mitigate blame when it provides a worse offer. Potential ethical concerns with the use of AI to bypass consumer resistance to negative offers are discussed.

Keywords

artificial intelligence, anthropomorphism, algorithm, robots, technology, pricing, intentions, expectations, satisfaction

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Marketing managers currently find themselves in a period of technological transition, wherein artificial intelligence (AI) agents are increasingly viable replacement options for human representatives in administering product and service offers directly to customers. AI agents have been adopted across a broad range of consumer domains to handle both face-to-face and remote customer transactions (Davenport et al. 2020; Harris, Kimson, and Schwedel 2018; Huang and Rust 2018; Wirtz et al. 2018), ranging from traditional retail and travel (Mende et al. 2019) to ride and residence sharing (Hughes et al. 2019) and even legal and medical services (Esteva et al. 2017; Turner 2016). Given AI agents' advanced information processing capabilities and labor cost advantages (Kumar et al. 2016), the transition away from human representatives in administering product and service offers seems predominately advantageous for firms. Despite AI's potential, scant research has examined how consumers evaluate AI systems in relation to equivalent offerings delivered by humans.

The increasingly pervasive use of AI raises the possibility that offers administered by AI agents may impact consumer response in novel ways as compared with offers administered by human agents. For example, Uber uses an AI machine learning system to estimate travel elasticities and administer ride price offers (Newcomer 2017). Imagine a customer who pays for an Uber ride downtown but, for the return trip, unexpectedly receives a price offer triple the original. Does administration of sthis worse-than-expected offer by an AI agent, rather than a human agent, have implications for purchase likelihood, customer satisfaction, or intentions toward the use of Uber in the future? What if the return trip was unexpectedly much cheaper than anticipated, resulting in a much better-than-expected offer for the consumer? The present research theorizes and demonstrates an interaction between the type of agent and outcome expectation discrepancies, such that consumers respond less

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negatively to worse-than-expected price offers when transmitted by AI agents (compared with human agents) and more positively to better-than-expected price offers when transmitted by human agents (compared with AI agents). In addition, this interaction is further moderated by anthropomorphic characteristics of the AI. The more humanlike the AI is in terms of its appearance or cognitive functions, the more it reduces the expectations discrepancy gap between human and AI agents. This moderator represents a strategic managerial input that can be used to manage customer satisfaction on the basis of the customer's price expectations.

Our theory and findings have key implications for firms enlisting both human and AI agents who administer outcomes that are discrepant from expectations. These findings are relevant for price offers, in addition to other situations where consumers learn of unexpectedly negative outcomes along dimensions other than price, such as cancellations, delays, negative evaluations, status changes, product defects, rejections, service failures, and stockouts. Our findings are also pertinent to instances where consumers receive unexpectedly positive outcomes such as expedited deliveries, rebates, upgrades, service bundles, exclusive offers, loyalty rewards, and customer promotions. Managers can apply our findings to prioritize (vs. postpone) human-to-AI role transitions in situations where negative (vs. positive) discrepancies are more frequent and impactful. Moreover, our results suggest that even when a role transition is not holistically passed to an AI, the selective recruitment of an AI agent to disclose certain discrepant information can still be advantageous. Firms that have already transitioned to consumer-facing AI agents, including the multitude of online and mobile applications that use AI-based algorithms to create and administer offers, also stand to benefit from our findings. Our research reveals that AI agents should be selectively anthropomorphized (i.e., depicted as either machinelike vs. humanlike) depending on whether an offer will be worse or better than expected.

Our research contributes to the literature in multiple ways. First, we show that AI (vs. human) agents asymmetrically alter the effects of outcome expectation discrepancies on purchase, satisfaction, and reengagement intentions, thereby broadening the associated literature examining human–AI transactions in marketing contexts (e.g., Huang and Rust 2018; Kim and Duhachek 2020; Longoni, Bonezzi, and Morewedge 2019; Mende et al. 2019; Srinivasan and Abi 2021). Whereas the growing literature on technology in marketing has shown that consumer engagement tends to decrease when an AI agent administers a transaction (e.g., through a perceived lack of human mental or emotional attributes on the part of the representative; Longoni, Bonezzi, and Morewedge 2019; see also Dietvorst, Simmons, and Massey 2015), we reveal how transactions with AI can either improve or undermine purchase tendencies and reengagement intentions depending on the valence of deviation from consumer expectations.

Second, whereas previous research has shown largely positive consequences of anthropomorphized products in the form of increased consumer engagement and product liking (Aaker, Vohs, and Mogilner 2010; Aggarwal and McGill 2007; Waytz, Heafner, and Epley 2014), or generally negative compensatory responses (Mende et al. 2019), we show an asymmetric pattern such that consumers transacting with anthropomorphized AIs are more likely to react negatively to worse-than-expected marketing offers but respond more favorably to better-than-expected offers from anthropomorphized AIs owing to the role of perceived intentions. Third, we provide evidence that our effects are not driven by potential alternative explanations, including uncanny valley theory (Mori, MacDorman, and Kageki 2012) or AIs' superior market tracking ability. We note that prior research exploring the anthropomorphism of AI agents has argued for an uncanny valley effect that leads people to avoid agents that appear too humanlike, ostensibly due to eeriness that emerges after a certain inflection point (Kim, Schmitt, and Thalmann 2019; Mende et al. 2019; Mori, MacDorman, and Kageki 2012). Our research demonstrates AI anthropomorphism patterns that are ostensibly incompatible with an uncanniness explanation and that emerge even when empirically controlling for uncanniness perceptions. Instead, we reveal how AI (vs. human) agents are inferred to have weaker selfish and benevolent intentions when developing discrepant offers, with implications for subsequent consumer response. In doing so, we suggest a new and important process driving both aversion to and engagement with anthropomorphized AI agents.

As a fourth contribution, our research reveals that the use of AI agents to administer offers can simultaneously present ethical dilemmas alongside opportunities for marketing firms. On the one hand, our work reveals opportunities for firms to receive due (and perhaps otherwise overlooked) credit for offers that are better than expected, with corresponding improvements in customer response. On the other hand, our work also reveals the possibility of AI misuse as a darker tool through which marketers can increase the acceptance of offers that fall short of expectations while simultaneously increasing intentions to reengage with the offending firm.

Theoretical Framework

Expectations Discrepancy Theory in Marketing Transactions

We examine the impact of AI versus human product and service offer administration through the theoretical lens of discrepant outcome expectations—that is, consumer reactions to offers that are better or worse than expected on a key dimension. Although expectation discrepancies have been examined in a variety of interpersonal sales transaction contexts (e.g., Darke, Ashworth, and Main 2010; Evangelidis and Van Osselaer 2018; Oliver, Balakrishnan, and Barry 1994), to our knowledge, no studies have investigated the implications of AI agents as transaction administrators. Although no research has specifically examined this phenomenon, extant research into robotics and artificial intelligence suggests that AI (vs. human) agents could have either positive or negative effects when administering price offers that are better or worse than expected. We next

consider the extant literature on expectancy theory, followed by a conceptual integration with recent AI research.

The broad body of research that examines expectation discrepancies for product and service offers assumes that a set of expectations exists prior to entering a consumption event, and offers trigger a comparison against expected referents (Evangelidis and Van Osselaer 2018; Oliver and DeSarbo 1988). For example, a buyer entering into a potential Uber transaction could have expectations for a price of $20 based on historical experience but then receive a ride price offer of $30 (worse than expected) or $10 (better than expected). Worse-than-expected offers elicit an adverse reaction and have been demonstrated to lead to lower purchase likelihood, satisfaction, and desire to reengage with specific sales agents, whereas better-than-expected offers produce favorable consumer responses (Oliver, Balakrishnan, and Barry 1994; Spreng, MacKenzie, and Olshavsky 1996). These expectations can relate to many aspects of service and product experiences, including temporal dimensions such as product/service delivery time, attribute performance, price, and so on. Any dimension of consumer experience for which outcomes deviate from experience can potentially be used to form evaluations of the agent.

The current research aims to conjoin the extant literature on technology and expectations discrepancy theory by highlighting the important role of AI. The previous literature on expectations discrepancy has assumed a human agent during offer administration. AI is perceived differently than human agents in many aspects because it is an artificially created technology (Dietvorst, Simmons, and Massey 2015; Longoni, Bonezzi, and Morewedge 2019; Mende et al. 2019). Moreover, AI technology endowed with machinelike versus humanlike features (e.g., a name or face) appears to trigger different perceptions about its capabilities (MacDorman, Vasudevan, and Ho 2009; Waytz, Heafner, and Epley 2014), thereby potentially altering responses to offers that are discrepant from expectations.

Transactions with AI Versus Human Agents: Inferred Intentions Matter

Previous research has shown that human agents are more liked and trusted than AI agents across a variety of contexts (Dietvorst, Simmons, and Massey 2015; Longoni, Bonezzi, and Morewedge 2019). For example, consumers are more likely to prefer a medical service provided by a human versus an AI agent because AI agents are considered incapable of considering each patient's unique, individual, and situational characteristics (Longoni, Bonezzi, and Morewedge 2019). Thus, one could predict that the administration of an offer, such as offering a price for an Uber trip, by an AI (vs. a human) agent could negatively influence purchase, satisfaction, and reengagement outcomes regardless of how well the offer matches expectations. Moreover, extant research has shown that people have less aversion to inflicting physical punishment on an embodied AI (i.e., a robot) than a human (Złotowski et al. 2015), suggesting in particular that a worse-than-expected offer elicited by an AI could be met with a more adverse customer response. However, extant research has not considered transactional contexts that vary in the valence of outcome expectation discrepancy, for which we theorize that a different pattern of effects will emerge. We extend previous research revealing an AI aversion phenomenon in the contexts of prediction tasks and medical decisions (Dietvorst, Simmons, and Massey 2015; Longoni, Bonezzi, and Morewedge 2019). In doing so, we contrast AI and human agents in the dimension of intentions inferred from their actions and theorize that transactional offers with the same face value could be evaluated as more acceptable when provided by an AI (vs. a human) agent, depending on the valence of the offer.

According to established models of intentionality, an action is perceived to be driven by intentions when it involves three key elements: a desired outcome based on one's own motives, a belief that the action will lead to that outcome, and one's autonomous decision to take the action (Brand 1984; Bratman 1987; Malle, Moses, and Baldwin 2001). Research has shown that individuals are capable of inferring intentions from others' actions, and an action that suffices for all three elements is perceived to have stronger intentions. A growing body of research suggests that AI (vs. human) agents are considered to have a lower capacity for self-motivated decision making and, by extension, lack the capacity to form their own intentions that drive subsequent actions. Because an AI agent is a nonhuman machine made by humans to serve humans (Russell and Norvig 2010), lay beliefs have formed such that AI agents' actions serve human goals and fulfill extrinsic human desires (Huang and Chen 2019; Kim and Duhachek 2020; Russell and Norvig 2010). Put differently using intentionality theory, AI agents lack the primary element of intentionality, namely self-motivated "desired outcomes," and thus, they do not qualify as agents that possess their own intentions (Brand 1984; Bratman 1987; Malle, Moses, and Baldwin 2001).

Social information processing theory has revealed that when approaching an interpersonal engagement with meaningful outcomes—such as a marketing transaction—individuals attempt to judge the other party primarily on inferred intentions (Abele and Wojciszke 2007; Wojciszke, Abele, and Baryla 2009). Of particular importance to determining response is whether the intentions driving the other party's decision are inferred to be selfish (i.e., placing the self first) or benevolent (i.e., considerate of others; Wojciszke, Abele, and Baryla 2009). Inferred selfish (benevolent) intentions decrease (increase) favorable evaluation of an agent's actions. For example, imagine that a human driver of a vehicle swerved into a wall to protect a child who jumped into the street chasing a ball. The benevolence of the action, if conducted by a human driver, would be praised, but the same action would receive less praise when conducted by an autonomous vehicle's AI system (Awad et al. 2018; Gray, Gray, and Wegner 2007; Gray, Young, and Waytz 2012). This explanation is also consistent with research demonstrating weaker perceptions of goodwill from benevolent actions conducted by humans (e.g.,

corporate social responsibility, prosocial behaviors) when those actions are not self-motivated (Barasch et al. 2014; Bolton and Mattila 2015; Huang and Chen 2019; Du, Bhattacharya, and Sen 2007; Rand, Newman, and Wurzbacher 2015). Moreover, inferences that a human agent's better-than-expected offer is driven by benevolent intentions tends to improve the recipient's evaluation of that offer (Hilbig et al. 2015; Radke, Güroğlu, and De Bruijn 2012).

Conversely, we propose that a worse-than-expected decision driven by ostensibly selfish intentions (e.g., hitting the child instead of swerving into a wall) would be criticized more when conducted by a human (vs. an AI) agent. Such judgment is echoed across multiple contexts indicating that intentional violations by humans toward others are perceived to be more egregious than unintentional violations (e.g., in sports, criminal behavior, and resource management; Gray 2012; Güroğlu, Van den Bos, and Crone 2009). For example, electric shocks are less painful when administered unintentionally compared with when administered with a malicious intent (Gray 2012). Intentions are also codified in criminal justice systems, such that acts committed while lacking intent of malice or selfishness are often met with milder punishment and individuals often receive less blame for their actions. A similar pattern emerges in marketing contexts for human agents that administer worse-than-expected offers, which are evaluated more negatively if inferred intentions are selfish, but less so if unintentional (Tsiros, Mittal, and Ross 2004). Extending this to our present research, and based on the premise that AI lacks the capacity for intentions, we propose that consumers will be less likely to infer benevolent intentions (in the case of a better-than-expected offer) and selfish intentions (in the case of a worse-than-expected offer) for the same offer transaction administered by an AI agent versus a human agent.

We theorize that this difference in inferred intentions will impact consumer purchase responses such as purchase likelihood and satisfaction. In the case of a worse-than-expected offer (vs. a better-than-expected offer), we propose that a human agent's intentions will be inferred as driven more by selfish intentions (vs. benevolent intentions), which, in turn, reduces (vs. improves) the likelihood of offer acceptance and subsequent customer satisfaction.

Stated formally,

H_1: The effect of a price offer that is worse (better) than expected in lowering (raising) purchase likelihood and satisfaction is weakened when administered by an AI agent versus a human agent.

H_2: The effect proposed in H_1 is mediated by the inferred intentions of the offering agent. Specifically, in the case of a better-than-expected (vs. worse-than-expected) price offer, greater inferred benevolent (vs. selfish) intentions improve (vs. undermine) offer response. Both mediating intention pathways are attenuated when the offer is administered by an AI agent versus a human agent.[1]

Empirical Overview

We evaluate our theory over a series of five studies. In Studies 1a and 1b, we examine our basic effect for worse-than-expected (Study 1a) and better-than-expected (Study 1b) offers to test H_1. Study 2 tests our full model, including underlying process mechanisms, thereby testing H_1 and H_2. Studies 3a and 3b introduce a third hypothesis on the moderating effect of anthropomorphizing AI agents.

Study 1a: Response to a Worse-Than-Expected Offer from an AI Agent Versus a Human Agent

The primary objective of Study 1a was to test our proposition that acceptance of a worse-than-expected offer will be higher when administered by an AI agent versus a human agent. This study utilized a product resale context (resale of an aftermarket concert ticket) in which participants imagined receiving a price offer that was administered by either an AI agent or a human agent.

Method

Participants and design. The experiment was a 2 (agent type: human, AI) × 2 (offer type: worse than expected, expected) group, between-subjects design. A total of 174 undergraduate students ($M_{age} = 21.2$ years; 56% female) participated in this study in return for course credit. We referred to previous AI research with similar experimental designs and determined our sample size based on the effect sizes and samples sizes reported in these studies (Longoni, Bonezzi, and Morewedge 2019; Mende et al. 2019). The sample size for this and subsequent studies was determined to provide sufficient power to detect a medium size effect if it exists (Bausell and Li 2002). In addition, sample size was influenced by the size of the participant pool made available to the authors in the given semester.

Procedure. Participants were first instructed to provide their favorite musician whom they would most like to see in concert. Next, participants read a scenario in which they were asked to imagine that all the tickets for an upcoming concert of their favorite musician were sold out and the only way to attend the concert was to buy a resale ticket through an online ticket resale website. Participants in the human (AI) agent condition were told that the ticket was being sold by "another person" ("an artificial intelligence"). All participants were told that the agent was providing a price offer of $140. Offer type was manipulated by telling the participants that the ticket was sold either at the same or a lower price to a different customer. Participants in the worse-than-expected (expected) offer condition were told that a similar ticket was recently sold for $110 ($140) to someone else. This approach of disclosing a lower sale price for the product to a different consumer is consistent

[1] For an illustration of the model proposed by H_2, see Web Appendix A.

with long-standing practice for eliciting a negative expectation discrepancy (e.g., Fisk and Young 1985). The offer type manipulation was validated with a pretest that showed a significant difference in price expectations between the two conditions (the pretest results are presented in the next section; further details on the stimuli and pretest are available in Web Appendix B). Then, participants indicated whether they would purchase the ticket (yes/no). Finally, participants responded to background questions (e.g., gender).

Results

Pretest. We conducted a pretest of offer type to assess our operationalizations of worse than expected (ticket sold to someone else for $110; participants were offered $140) and expected (ticket sold to someone else for $140; participants were offered $140). One hundred fifty respondents from Amazon Mechanical Turk (MTurk) completed the pretest in return for monetary compensation. After reading the ticket scenario without agent descriptions, respondents answered the two-item expectancy disconfirmation scale from Oliver (1980) adapted to this context: "Think about what your expectations were for the price offer. How does the price that you were offered compare to your expectations?" (1 = "I expected a much lower price," 4 = "Price was as I expected," and 7 = "I expected a much higher price"; 1 = "Much worse offer than I expected," 4 = "Offer was as I expected," 7 = "Much better offer than I expected"). Analysis of the two-item composite (r = .80) revealed that the worse-than-expected offer condition was significantly below both the "as expected" midpoint (i.e., 4) ($M_{worse} = 2.90$, SD = 1.36; $t(75) = -7.06$; $p < .001$) and the expected condition ($M_{expected} = 3.98$, SD = .96; $F(1, 148) = 31.52$, $p < .001$, $\eta_p^2 = .18$), indicating a successful manipulation.

Offer acceptance. First, offer type (0 = worse, 1 = expected), agent type (0 = AI, 1 = human), and offer acceptance (1 = yes, 0 = no) were contrast coded and submitted to a binomial logistic regression. We observed a significant main effect of agent ($\chi^2 = 7.92$, $p = .005$) and no main effect of offer type ($\chi^2 = .40$, $p = .53$), both subsumed by a significant two-way interaction ($\chi^2 = 6.92$, $p = .009$). Further chi-square analysis revealed that the worse-than-expected offer was accepted more in the AI condition (49%) than in the human condition (19%; $\chi^2 = 8.39$, $p = .004$), thus confirming H_1. In the expected offer condition, there was no effect of agent on offer acceptance ($\chi^2 = .60$, $p = .44$; see Figure 1).

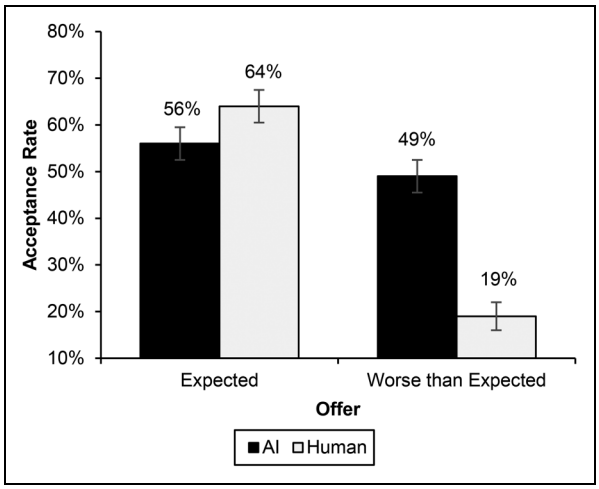


Figure 1. Offer Acceptance Rate as a Function of Offer and Agent Type (Study 1a).
Notes: Error bars = ±1 SEs.

Discussion

Consistent with our theory and H_1, the results of Study 1a indicate that consumers are more likely to accept a worse-than-expected product offer from an AI agent versus a human agent. This result is evidence that consumers respond differently to AI and human agents with respect to price expectation discrepancies stemming from marketing transactions. Our theory proposes that a better-than-expected offer will also provide a different pattern of results for offers administered by an AI agent versus a human agent. In Study 1b, we examine whether better-than-expected offers elicit a different pattern of response when administered by AI versus a human agent.

Study 1b: Response to a Better-Than-Expected Offer from an AI Agent Versus Human Agent

Study 1b addresses the case of a better-than-expected offer. We posit that offer administration by a human leads to more positive consumer outcomes as compared with AI.

Method

Participants and design. Two hundred ninety-nine MTurk participants ($M_{age} = 41.9$ years; female = 53%) completed this study for monetary compensation. All participants were randomly assigned to one of the conditions in a 2 (agent type: AI, human) × 2 (offer type: expected, better than expected) between-subjects design.

Procedure. The procedure was a ticket resale context adapted from Study 1a, with the exception that the worse-than-expected offer was replaced with a better-than-expected offer (for stimuli and pretest details, see Web Appendix C). Specifically, subjects in the better-than-expected condition were told that a similar ticket was sold recently for $170 and were offered a price of $140. Subjects in the expected condition were told that the ticket recently sold for $140 and were offered a price of $140. After receiving the offer, participants indicated whether they would purchase the ticket (yes/no), then responded to background questions (e.g., gender).

Results

Pretests. As in Study 1a, we conducted a separate pretest to validate our offer type manipulation with 150 MTurk participants who received monetary compensation. Using the same two items from Study 1a, we measured the extent to which participants perceived that the offer was better than they expected. Analysis of the two-item composite (r = .72) revealed that the mean score of the composite was higher in the better-than-expected offer condition ($M_{better} = 5.83$, SD = 1.25) when compared with the expected offer condition ($M_{expected} = 4.18$, SD = 1.00; $F(1, 148) = 80.8$, $p < .001$, $\eta_p^2 = .35$), indicating a successful manipulation.

Offer acceptance. First, offer type (0 = expected, 1 = better), agent type (0 = AI, 1 = human), and offer acceptance (1 = yes, 0 = no) were contrast coded and submitted to a binomial logistic regression. We observed a significant main effect of agent ($\chi^2 = 5.94$, $p = .02$) but no significant main effect of offer type ($\chi^2 = .99$, $p = .32$), both subsumed by a significant two-way interaction ($\chi^2 = 4.79$, $p = .03$). Further chi-square analysis revealed that the better-than-expected offer was accepted more in the human condition (89%) than in the AI condition (76%; $\chi^2 = 6.30$, $p = .01$), thus confirming H_1. In the expected offer condition, there was no effect of agent on offer acceptance ($\chi^2 = .17$, $p = .69$; see Figure 2). This pattern supports H_1.

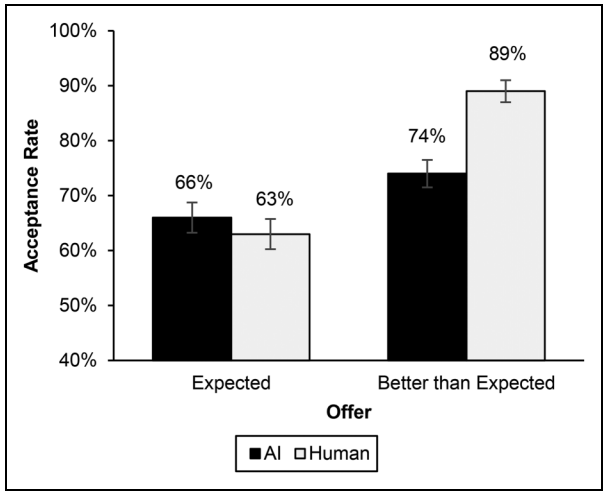


Figure 2. Acceptance of Better-Than-Expected Offers by Agent Type (Study 1b).
Notes: Error bars = ±1 SEs.

Discussion

Study 1b demonstrated that consumer response to better-than-expected offers are systematically influenced by the administrating agent. Consistent with our theory, we found that the same better-than-expected offer elicits a more positive evaluation when administered by a human instead of an AI agent. Coupled with the results of Study 1a, these findings provide direct support for H_1.

Study 2: Inferred AI Intentions Alter Offer Acceptance

Study 2 served multiple objectives. First, in this study we build on Studies 1a and 1b to test our full model by simultaneously considering both worse-than-expected and better-than-expected offers. Second, we assess our proposed process model, thereby testing H_2. Specifically, we investigate whether the intentions of an AI (vs. a human) agent are inferred to be less selfish (in the case of a worse-than-expected offer) and less benevolent (in the case of a better-than-expected offer), which in turn influences offer evaluation. Study 2 also tests for several alternative explanations, including uncanniness. Study 2 employs a ride-sharing service context to build on the results of Study 1b.

Method

Participants and design. All participants were randomly assigned in a 2 (agent type: human, AI) × 3 (offer type: worse than expected, expected, better than expected) between-subjects design. Six-hundred ninety-eight undergraduate business students ($M_{age} = 21.2$ years; female = 59%) completed this study for partial course credit.

Procedure. Participants first read an introduction, which explained that they would participate in a study containing a scenario related to Uber, a ride-sharing service. First, participants read a short passage explaining that the price offered at Uber is determined by an Uber agent, with the description of that agent manipulated between conditions. We adapted agent type descriptions from previous research on AI and service robots (Kim and Duhachek 2020; Mende et al. 2019). Participants were told, "This is your Uber agent who will personally determine the offer price of your Uber rides, named [Alex/XT-1000]." In the human condition, an image of a human was presented, adapted from Mende et al. (2019), whereas in the AI condition an image of a robot pretested to be moderately humanlike was presented (stimuli details in Web Appendix D; Development procedures for AI stimuli in Web Appendix G).

Next, participants read about and evaluated a price offer from the Uber agent. The scenario explained that the initial cost of a ride from home to a restaurant for dinner was $20, and for the return trip home they were later offered the price of $20 in the expected condition, $30 in the worse-than-expected condition, and $10 in the better-than-expected condition. As the dependent measure, we asked participants their likelihood of accepting the offer (1 = "not at all likely," and 7 = "very likely").

To assess process, we collected measures of inferred agent intentions when administering the offer. The measurement items for the selfish and benevolent intentions were developed based on previous studies of valenced intentionality (Gray 2012; Gray and Wegner 2008). Specifically, we collected three items measuring inferred benevolent intentions ("To

what extent do you agree with the following statements about Uber agent's own, personally created intention when forming your individual ride price offer? Uber agent had their own …" "benevolent intentions," "generous intentions," "good intentions" [1 = "strongly disagree," and 7 = strongly agree"]; the three items were averaged to create an index of benevolent intentions, $\alpha = .84$) and three items measuring inferred selfish intentions ("Uber agent had their own …" "selfish intentions," "greedy intentions," "bad intentions" [1 = "Strongly disagree," and 7 = "Strongly agree"] averaged to create an index of selfish intentions, $\alpha = .93$).

We also recorded measures for alternative processes, including uncanniness ("I feel uneasy," "I feel unnerved," "I feel creeped out"), whether respondents may have perceived the original price to the restaurant to be overpriced ("Do you feel that you overpaid for the first Uber trip that you took to the restaurant?"), whether respondents perceived different price tracking capacity between the two agents ("To what extent do you think the Uber agent was able to track the real-time price changes of other people's rides and made an offer to you based on this market information?"), and whether respondents perceived that the offer is controlled by someone else ("To what extent do you think that the Uber agent's price offer was controlled by another person behind the scene?"). Finally, participants responded to background questions (e.g., gender).

Results

Pretests. Just as in the previous studies, we conducted separate pretests to validate our manipulations of agent type and offer type (for details, see Web Appendix D). The pretest of offer type assessed our operationalizations of a worse-than-expected offer ($30 price offer with a $20 precedent), better-than-expected offer ($10 price offer with a $20 precedent), and the expected price offer ($20 price offer with a $20 precedent). Three hundred respondents from MTurk completed the pretest in return for monetary compensation. Respondents answered the same two-item expectation discrepancy scale used in previous studies. Analysis of the two-item composite ($r = .81$) revealed the main effect of offer type ($F(2, 297) = 82.85$, $p < .001$, $\eta_p^2 = .36$). Additional contrasts revealed that the mean score of the composite was significantly higher in the better-than-expected offer condition ($M_{better} = 5.19$, $SD = 1.47$) compared with the expected offer condition ($M_{expected} = 3.99$, $SD = .94$; $F(2, 297) = 46.05$, $p < .001$, $\eta_p^2 = .19$), and the mean score of the composite in the expected offer was higher than the worse-than-expected offer condition ($M_{worse} = 2.84$, $SD = 1.40$; $F(2, 297) = 46.27$, $p < .001$, $\eta_p^2 = .19$). These results indicate a successful manipulation of offer type.

Offer acceptance likelihood. A 2 (agent type: human, AI) × 3 (offer type: worse than expected, expected, better than expected) analysis of variance (ANOVA) revealed a significant main effect of offer type ($F(2, 692) = 150.61$, $p < .001$, $\eta_p^2 = .30$) and no main effect of agent type ($F(2, 692) = .06$, $p = .80$, $\eta_p^2 < .01$), both subsumed by a significant two-way interaction ($F(2, 692) = 7.18$, $p = .001$, $\eta_p^2 = .02$). A series of planned contrasts revealed that acceptance of the expected offer was not significantly different between agent types ($M_{human} = 3.69$, $SD = 1.99$; $M_{AI} = 3.40$, $SD = 1.75$; $F(2, 692) = 1.71$, $p = .19$, $\eta_p^2 = .006$). In contrast, and as we predicted, acceptance of the better-than-expected offer was significantly higher in the human (vs. AI) condition ($M_{human} = 5.58$, $SD = 1.19$; $M_{AI} = 5.23$, $SD = 1.62$; $F(2, 692) = 4.28$, $p = .04$, $\eta_p^2 = .02$). Also as predicted, acceptance of the worse-than-expected offer was significantly higher in the AI (vs. human) condition ($M_{human} = 2.53$, $SD = 1.59$; $M_{AI} = 3.17$, $SD = 1.90$; $F(2, 692) = 8.46$, $p = .004$, $\eta_p^2 = .03$) (see Figure 3).

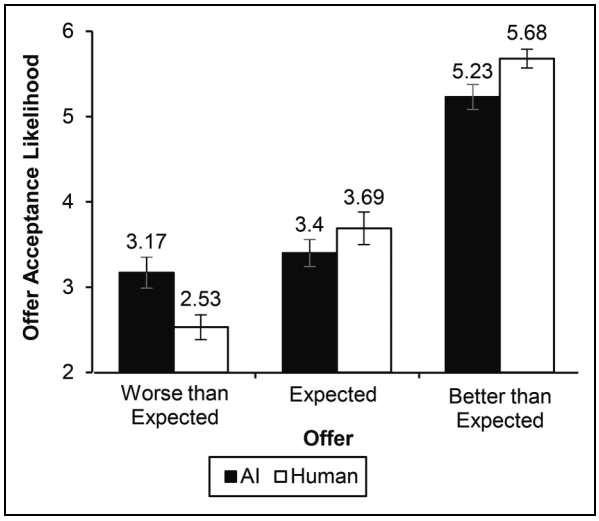


Figure 3. Offer Acceptance Likelihood as a Function of Offer and Agent Type (Study 2).
Notes: Error bars = ±1 SEs.

Moderated mediation via intentions. To assess the underlying process, we first evaluated selfish and benevolent intentions via ANOVA. The 2 (agent type: human, AI) × 3 (offer type: worse than expected, expected, better than expected) ANOVA of selfish intentions revealed a significant main effect of offer type ($F(2, 692) = 18.39$, $p < .001$, $\eta_p^2 = .05$) and no main effect of agent type ($F(2, 692) = 1.95$, $p = .16$, $\eta_p^2 = .003$), and the predicted two-way interaction matched the results for offer acceptance ($F(2, 692) = 7.60$, $p = .001$, $\eta_p^2 = .02$; for the profile plot and pairwise comparisons, see Web Appendix D). The 2 × 3 ANOVA of the benevolent intentions measure revealed a significant main effect of offer type ($F(2, 692) = 20.53$, $p < .001$, $\eta_p^2 = .06$) and a marginally significant main effect of agent type ($F(2, 692) = 3.64$, $p = .057$, $\eta_p^2 = .01$), both subsumed by the predicted two-way interaction ($F(2, 692) = 9.93$, $p < .001$, $\eta_p^2 = .03$; for profile plots and pairwise comparisons, see Web Appendix D, p. 9).

To assess mediation via intentions, we conducted a moderated mediation analysis using a Hayes (2017) PROCESS model (Model 7) in which offer type (worse than expected, expected, better than expected) served as the multicategorical independent variable, inferred intentions (i.e., selfish and benevolent intention perceptions) served as the two mediating

variables, offer acceptance likelihood served as the dependent variable, and agent type (human, AI) served as the moderating variable. This analysis revealed that the index of moderated mediation for both selfish and benevolent intentions did not include 0 (indirect effect for selfish intention = .12; 95% confidence interval: [.05, .21]; indirect effect for benevolent intention = .14; 95% confidence interval: [.07, .23]), indicating that the offer type → inferred intentions → acceptance likelihood path is moderated by agent type.

Alternative mechanisms. The first alternative mechanism assessed was the uncanniness index (three-item average, $\alpha = .91$). An ANOVA of the uncanniness index revealed that AI was perceived as more uncanny than a human only in the expected offer condition ($M_{AI} = 2.98$, $SD = 1.29$; $M_{human} = 2.50$, $SD = 1.35$; $F(2, 692) = 6.96$, $p = .01$). More germane to our inquiry, the two-way interaction between the agent and offer type was not significant, resulting in a pattern inconsistent with explaining our results. The second alternative mechanism assessed was whether the perception the original fare (i.e., \$20 to get to the restaurant) was overpriced. An ANOVA of the overpricing perception revealed a significant main effect of offer type ($M_{worse} = 4.14$, $SD = 1.70$; $M_{expected} = 5.10$, $SD = 1.53$; $M_{better} = 5.52$, $SD = 1.39$; $F(2, 692) = 49.00$, $p < .001$). However, there was no interaction between the agent and the offer ($F(2, 692) = 2.19$, $p = .11$), precluding it as a driver of the observed pattern of mediation. Third, we assessed the agent's market-price-tracking capability as an alternative mechanism. An ANOVA of price tracking capability revealed a significant main effect of agent ($F(2, 692) = 22.90$, $p < .001$) and offer type ($F(2, 692) = 3.38$, $p = .04$). However, there was no interaction between the agent and the offer ($F(2, 692) = .16$, $p = .86$), resulting in a pattern inconsistent with explaining our results. Finally, ANOVA of the perception that the offer is controlled by another person revealed only a significant main effect of agent ($F(2, 692) = 6.03$, $p = .01$). The main effect of offer ($F(2, 692) = 2.03$, $p = .13$) and the interaction ($F(2, 692) = 1.60$, $p = .20$) were not significant, also resulting in a pattern inconsistent with explaining our results. Taken together, these results rule out multiple alternative explanations and lend further confidence to our conceptual model.

Discussion

The results of Study 2 extend our understanding of how offers that are discrepant from expectations systematically elicit different responses between human and AI agents. We replicate the differential human versus AI effects observed in Studies 1a and 1b, supporting H_1. Specifically, worse-than-expected offers are more likely to be accepted when administered by an AI, whereas better-than-expected offers are more likely to be accepted when administered by a human. Moreover, our results reveal that inferred intentions mediate the observed effects, in support of H_2. Specifically, a worse-than-expected (better-than-expected) offer from a human increases (decreases) inferred selfish intentions and decreases (increases) inferred benevolent intentions, which in turn decreases (increases) offer acceptance. However, an AI agent attenuates these pathways, thereby leading to an acceptance increase in the case of a worse-than-expected offer, but an acceptance decrease in the case of a better-than-expected offer, compared with a human agent. In addition, this study rules out several potential alternative process mechanisms.

Studies 3a and 3b: Anthropomorphism of AI Agents

In our final two studies, Studies 3a and 3b, we extend our work beyond the human–AI dichotomy to consider different forms of AI agents. We propose that AI agents may be anthropomorphized through their presentation to be perceived as more humanlike (vs. machinelike), thereby eliciting patterns more consistent with human agents. In doing so, we provide insights for managers on how to best depict their AI agents (i.e., as more humanlike vs. more machinelike) in situations where consumers will receive better- or worse-than-expected offers.

Research has revealed that endowing an AI with humanlike features typically makes people perceive the AI more positively. For example, consumers assessing an AI agent that controlled an autonomous vehicle became more trusting as that agent was presented as increasingly humanlike, rather than machinelike (Waytz, Heafner, and Epley 2014). Indeed, the marketing literature has shown that anthropomorphized products are predominately liked more and improve consumer engagement. For example, thinking about a product (e.g., a car) through an anthropomorphic perspective increases affection toward that product and leads to less willingness to replace it (Chandler and Schwarz 2010). Research has also shown that a product that resembles a human face in its design could increase perceived friendliness, leading to a positive evaluation of the product (Landwehr, McGill, and Herrmann 2011). As another instance, socially excluded consumers prefer anthropomorphized brands due to their need to establish relationships (Chen, Peng, and Levy 2017).

However, prior research investigating anthropomorphic AI has predominately focused on nonpurchase situations in which AI agents have been in a supporting role to consumers, such as providing safe transportation or providing medical advice. Our present inquiry builds on previous research by examining AI agents in a more adversarial role as administrators of product and service offers—that is, the consumer is sitting across the transaction table from the AI agent. In such transactional contexts, we propose that AI anthropomorphism will not have the generally favorable effect observed in many other contexts but will instead systematically either improve or worsen the impact of offers that differ from expectations.

The influence of AI anthropomorphism on response to discrepant offers will be driven by changes in the extent to which AI is perceived to be capable of intentionality (i.e., as defined by the previously discussed three elements of intentionality models; Brand 1984; Bratman 1987; Malle, Moses, and

Baldwin 2001), thereby altering the strength of inferred AI intentions. As an AI agent's humanlikeness increases, so too should its perceived capacity for intentionality. Indeed, a mind capable of intentionality has been argued to be a defining aspect of what it means to be uniquely human, and items specifically measuring related constructs are embedded in multiple established anthropomorphism scales (e.g., Kim and McGill 2011; Waytz, Cacioppo, and Epley 2010). Extant research has revealed that individuals are surprisingly willing to attribute advanced humanlike capabilities to anthropomorphized AI agents (Złotowski et al. 2015), potentially due in part to an overestimation of AI technical advancement stemming from fictional literature and film (Pollack 2006) and to the nature of human inference that assumes similar internal properties between entities that share common external appearances (Rozin and Nemeroff 2002). As such, AI agents that are viewed as increasingly anthropomorphic should be perceived to possess a greater capacity for intentionality, thereby generating stronger inferred intentions that increasingly elicit a response akin to a human agent.

We thus posit that anthropomorphizing an AI agent through the depiction of humanlike (vs. machinelike) properties will asymmetrically influence responses to discrepant offers. In the case of a better-than-expected offer, an anthropomorphized AI agent will lead to more positive downstream consequences (e.g., offer acceptance, satisfaction). Conversely, we hypothesize that the response to a worse-than-expected offer will become more negative when the AI agent is endowed with humanlike (vs. machinelike) properties. Stated formally,

> **H_3:** Anthropomorphism of an AI agent moderates the response to price offers that are better or worse than expected: a humanlike AI agent makes worse-than-expected offer responses more negative but makes better-than-expected offer responses more positive.

We test H_3 in Studies 3a and 3b. In doing so, we provide valuable insights to managers who are faced with decisions regarding customer interactions with AI agents. If H_3 is supported, our results suggest that selective anthropomorphism of AI agents could be applied to improve overall customer response to both worse- and better-than-expected offers.

Study 3a: The Moderating Role of Consumer Trait Anthropomorphism

In Study 3a, we extend our findings to consider the implications of anthropomorphism of AI agents for worse-than-expected offers. We do this utilizing an ecologically valid and managerially relevant segmentation measure: technology anthropomorphism (Waytz, Cacioppo, and Epley 2010). Technology anthropomorphism is the individual tendency to view technological products as possessing humanlike mental and emotional qualities. Consistent with our theory and H_1, we predict that consumers will be more likely to accept a worse-than-expected offer from an AI than a human. Moreover, in concordance with the role of anthropomorphism specified in H_3, this effect will be even stronger for individuals who perceive technological objects to be less humanlike, whereas the effect will attenuate for consumers who view technological objects as more humanlike. We also use a different context to test the robustness of our theory: an ultimatum game setting that examines the likelihood to accept both expected and unexpected allocations (Morewedge 2009; Thaler 1988). Prior research has demonstrated that the ultimatum game is a highly controllable context in which human offers may be contrasted with offers proposed by a nonhuman agent (e.g., Sanfey et al. 2003). We advance beyond previous research by examining the role of anthropomorphism of AI agents in the ultimatum context. Moreover, prior research has demonstrated that individuals have predictable and strong expectations regarding offer terms (Suleiman 1996), making this a context to which our theoretical predictions regarding worse-than-expected offers are particularly applicable.

Method

Participants and design. The experiment was a 2 (agent type: human, AI) × 2 (offer type: worse than expected, expected) between-subjects design. A total of 403 members of an MTurk online panel ($M_{age} = 34.9$ years; 64% female) participated in this study in return for financial compensation.

Procedure. Participants first completed the full Individual Differences in Anthropomorphism Questionnaire (IDAQ) scale for trait anthropomorphism of nonhuman agents (Waytz, Cacioppo, and Epley 2010), which includes a subscale specific to technological devices. The five-item technological anthropomorphism scale measures the extent to which technology is viewed as having humanlike mental and emotional qualities (e.g., "To what extent does the average computer have a mind of its own?," "To what extent does a car have free will?"; 0 = "not at all," and 10 = "very much"; for the full list of IDAQ items, see Web Appendix E). We averaged the five items to form an index of technology anthropomorphism ($\alpha = .85$). Then, participants were invited to participate in an ostensibly unrelated study on decision making. Participants were introduced to an ultimatum game in which the other player was either a "human" or an "artificially intelligent machine" who would administer an offer determining how $100 should be allocated. Previous research in behavioral economics has shown that an approximately even split (50%/50%) is both the modal offer by allocators and expectation by receivers, with splits even modestly outside of this range falling short of expectations and leading to an increasing tendency to reject the offer (Suleiman 1996). Thus, participants in the worse-than-expected offer condition received an offer of $10 to themselves and $90 to the other player, whereas participants in the expected offer condition received a $50/$50 offer. Participants then indicated their acceptance or rejection of the offer (1 = yes, 0 = no). Finally, participants responded to background questions (e.g., gender).

Results

Pretest. We conducted a pretest of offer type to assess our operationalizations of worse than expected ($10/$90) and expected ($50/$50). One hundred fifty respondents from MTurk completed the pretest in return for monetary compensation. After reading the ultimatum scenario, respondents answered an adaptation of the two-item expectations discrepancy scale as in the pretests for Studies 1a, 1b, and 2 (for pretest stimulus details, see Web Appendix E). Analysis of the two-item composite ($r = .92$) revealed that the worse-than-expected condition was significantly below the "as expected" midpoint ($M_{worse} = 1.68$, $SD = 1.21$; $t(72) = -16.36$; $p < .001$) and the expected condition ($M_{expected} = 4.10$, $SD = .59$; $F(1, 148) = 245.89$, $p < .001$, $\eta_p^2 = .18$), in support of a valid manipulation.

Offer acceptance. A binomial regression of offer acceptance (1 = yes, 0 = no) as a function of agent type (0 = human, 1 = AI) and offer type (0 = expected, 1 = worse than expected) revealed a main effect of agent type ($\chi^2 = 6.65$, $p = .01$) and main effect of offer type ($\chi^2 = 73.69$, $p < .001$), both subsumed by a significant two-way interaction ($\chi^2 = 4.00$, $p = .045$). In the case of an expected offer (i.e., an expected $50/$50 split), the offer was invariably accepted in both the AI (100%) and the human (100%) conditions, consistent with extant research in the ultimatum context. More germane to our theory, we conducted a chi-square analysis to compare the acceptance of worse-than-expected offers in the human and the AI conditions. As we predicted, acceptance of the worse-than-expected offer was significantly higher in the AI condition (78.6%) than in the human condition (60.4%; $\chi^2 = 7.73$, $p = .005$). These results of the purely human versus AI comparison were consistent with our theory and previous studies.

Moderation by trait anthropomorphism. We next tested how individual consumer tendencies to anthropomorphize AI (i.e., perceive AI as more human) altered offer acceptance. Consistent with H_3, we expected that individuals who view AI as not possessing humanlike characteristics would be more likely to accept a worse-than-expected offer from an AI, whereas those who do attribute humanlike characteristics would be less likely to accept a worse-than-expected offer. For expected offers, anthropomorphism index and agent type necessarily had no effect on offer acceptance, as expected offers were invariantly accepted at 100%. For worse-than-expected offers, we assessed the interaction between anthropomorphism index ($M_{anthro} = 2.01$, $SD = 1.57$) and agent type (human = 0, AI = 1) in determining acceptance through Johnson–Neyman (JN) analysis. Analysis revealed no main effect of anthropomorphism ($Z = 1.14$; $p = .26$) and a significant main effect of agent ($Z = 3.37$; $p = .001$), both subsumed by a significant interaction between the agent and anthropomorphism ($Z = -2.09$; $p = .037$). The result showed that for individuals with an anthropomorphism score below the JN point of 2.59—constituting 79.4% of the sample—acceptance of worse-than-expected offers was significantly higher from an AI versus a human. However, as anthropomorphism increased to the JN point of 2.59 and above—constituting 20.6% of the sample—no significant difference between agent type emerged (i.e., $p > .05$). No other regions of significance emerged at any higher levels of anthropomorphism (for JN graph and supplemental table illustrating interactions, see Web Appendix E, p. 14). That is, the positive effect of an AI (vs. human) agent on accepting a worse-than-expected offer emerges for individuals who are less likely to imbue the AI with humanlikeness, but does not hold for those individuals who increasingly attribute humanlike qualities to technology. These results support H_3.

Discussion

The results of Study 3a provide further evidence that worse-than-expected offers are more likely to be accepted when administered by an AI (vs. a human) agent, in support of H_1. Moreover, in support of H_3, individual perceptions of AI anthropomorphism alter this pattern of response; decreasing technology anthropomorphism (i.e., the AI is seen as less humanlike) increases the likelihood of accepting a worse-than-expected offer from an AI agent, whereas higher levels of technology anthropomorphism (i.e., the AI is seen as more humanlike) result in responses similar to those elicited by an actual human.

These results have several important implications. First, our findings provide managers with a new, easily measurable variable for segmentation when attempting to understand and predict how consumers will interact with their AI systems—individual tendency toward technology anthropomorphism. Second, our results suggest that anthropomorphism of the AI through means other than innate consumer tendencies, such as physical embodiment or descriptions of the AI mind, could impact response to unexpected offers. Indeed, the results of Study 3a reveal that most individuals tend to view technology as less than humanlike (e.g., the scale mean for technology anthropomorphism was substantially and significantly below the midpoint; $p < .001$), suggesting that the default view held by most consumers is that technology does not possess humanlike characteristics. In the next study, we examine how AI anthropomorphism can be manipulated, rather than measured, by marketers to lead to a differential acceptance of discrepant offers administered by AI.

Study 3b: Anthropomorphism, Satisfaction, and Reengagement

Study 3b expands on previous findings to examine how customer satisfaction and desire to maintain a relationship with the offering marketing agent varies depending on the level of humanlikeness of the AI agent administering a discrepant offer. Customer satisfaction is critical for firms (Anderson, Engledow, and Becker 1979; Williams and Naumann 2011), and evidence revealing how AI anthropomorphism alters the impact of (dis)satisfaction stemming from better- or worse-

than-expected offers would provide valuable insights for managers of AI agents. Regarding relationship maintenance, building a rapport between consumers and AI agents has become an important issue for companies utilizing AI agents as a primary contact point, with important implications for customer retention and relationship management (Huang and Rust 2018; see also Mende et al. 2019). As such, in this study, we provide participants a choice to maintain (vs. replace) the relationship with the current agent following a worse-than-expected offer. Consistent with our theory, we predicted that the effect of a better-than-expected offer on the tendency to maintain versus replace the relationship would be stronger for a humanlike (vs. machinelike) AI agent and that the opposite pattern would result in the case of a worse-than-expected offer. To assess this prediction, we selected a novel pair of AI agents differing only in their perceived humanlikeness (i.e., no differences in perceived uncanniness), based on a comprehensive study of multiple robots drawn from the ABOT (http://abotdatabase.info) database (development of AI stimuli detailed in Web Appendix G).

Method

Four hundred MTurk participants ($M_{age} = 36.7$ years, female = 48%) completed this study for monetary compensation. All participants were randomly assigned to one of the conditions in a 2 (agent type: machinelike AI, humanlike AI) × 2 (offer type: better than expected, worse than expected) between-subjects design. The overall procedure was similar to Study 2, which utilized an Uber ride-sharing context and explored customer satisfaction as the dependent variable. Participants read a short passage explaining that the price offered by Uber was determined by an AI agent (an initial $20 trip followed by a return trip price offer of $30 in the worse-than-expected condition and $10 in the better-than-expected condition). The images used to depict the two AI agents were drawn from the ABOT database (http://abotdatabase.info) and pretested regarding their humanlikeness and uncanniness (pretest reported in the study "Results" subsection, additional details in Web Appendices F and G).

Participants then answered three items related to offer satisfaction (identical measures to Study 1b; "satisfied," "appreciative," and "grateful" regarding the offer; 1 = "not at all," 7 = "very much"; the three items were averaged to create the index of satisfaction, $\alpha = .93$). We next asked participants the extent to which they wanted to interact with the same Uber agent in the future, which served as a measure of willingness to reengage with the agent. Participants were given an opportunity to stay with the current Uber agent in future interactions or discontinue the relationship with the current Uber agent and replace it with a different agent in a binary choice paradigm. The replacement agent stimulus was adopted from previous consumer research on AI (Mende et al. 2019; for the exact replacement choice stimuli, see Web Appendix F). Finally, participants answered background questions (e.g., gender).

Results

Pretests. Separate pretests supported the validity of our offer type and agent type manipulations. The pretest of agent type assessed the extent of perceived anthropomorphism (i.e., humanlikeness) for each agent. Using the same three-item humanlikeness ($\alpha = .86$) and three-item uncanniness ($\alpha = .89$) measures from our previous studies, a pretest of 100 MTurk respondents revealed that the humanlike (vs. machinelike) agent was perceived to be higher in humanlikeness ($M_{machinelike} = 1.65$, $SD = .83$; $M_{humanlike} = 2.47$, $SD = 1.07$; $F(1, 98) = 18.05$; $p < .001$) but not in uncanniness perceptions ($M_{machinelike} = 1.76$, $SD = 1.13$; $M_{humanlike} = 2.07$, $SD = 1.51$; $F(1, 98) = 1.35$; $p = .25$; additional pretest details and offer type pretest available in Web Appendix F).

Offer satisfaction. The satisfaction index was submitted to a 2 (agent type: machinelike AI, humanlike AI) × 2 (offer type: better than expected, worse than expected) ANOVA. The main effect of offer type was significant ($F(1, 396) = 301.19$, $p < .001$, $\eta_p^2 = .43$) and the main effect of agent was not significant ($F(1, 396) = .03$, $p = .87$, $\eta_p^2 < .01$). Consistent with our theory, we also found a significant interaction between agent and offer type ($F(1, 396) = 10.14$, $p = .002$, $\eta_p^2 = .03$). In the better-than-expected offer condition, satisfaction with the offer was higher in the humanlike agent condition ($M = 5.79$, $SD = 1.31$) than in the machinelike agent condition ($M = 5.29$, $SD = 1.21$; $F(1, 396) = 5.62$, $p = .02$, $\eta_p^2 = .02$). As predicted, we found the opposite pattern in the worse-than-expected offer condition. Satisfaction was higher in the machinelike agent condition ($M = 3.15$, $SD = 1.84$) than in the humanlike agent condition ($M = 2.69$, $SD = 1.59$; $F(1, 396) = 4.55$, $p = .03$, $\eta_p^2 = .02$, see Figure 4). This pattern is consistent with our theory, providing direct support for H_3.

Choice of agents. Offer type (0 = worse than expected, 1 = better than expected), agent (0 = machinelike, 1 = humanlike), and agent replacement decision (1 = replace, 0 = remain) were contrast coded and submitted to a binomial logistic regression. We found a significant main effect of offer type ($\chi^2 = 10.34$, $p = .001$) and a significant main effect of agent ($\chi^2 = 9.25$, $p = .002$), both subsumed by a significant interaction between agent and offer type ($\chi^2 = 12.41$, $p < .001$). Further chi-square directional tests were conducted to assess the simple effects of agent type at each level of offer type. A worse-than-expected offer led to a significantly higher proportion of replacement decisions in the humanlike agent condition (83%) than in the machinelike agent condition (63%) (one-sided test, $p = .001$; two-sided test, $p = .002$). In the better-than-expected offer condition, the proportion choosing replacement in the humanlike agent condition (28%) was lower than the machinelike agent condition (40%) (one-sided test, $p = .04$; two-sided test, $p = .07$). This pattern is consistent with our theory and demonstrates that humanlike AI is more (vs. less) likely to result in a continuation of the marketing relationship in the case of an offer that is better (vs. worse) than expected. Thus, we demonstrate that utilization of an appropriate AI

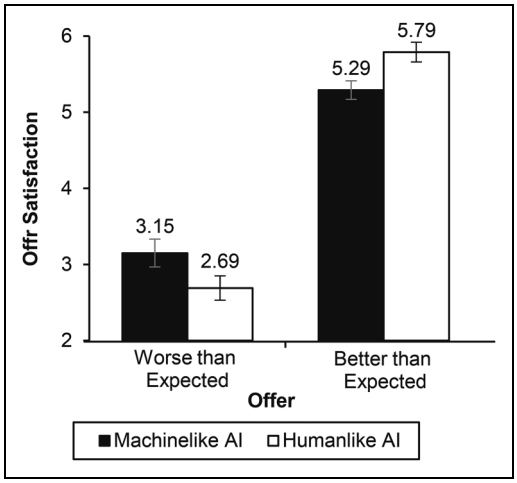


Figure 4. Offer Satisfaction as a Function of Offer and AI Agent Type (Study 3b).
Notes: Error bars = ±1 SEs.

agent leads to stronger relationships with that agent, which is a critical factor influencing customer retention and relationship management.

Discussion

Study 3b demonstrated that consumer response to better than expected and worse-than-expected offers are systematically influenced by the administrating AI agent. Consistent with our theory, we found that a better-than-expected offer elicits greater satisfaction and reengagement when administered by a humanlike (vs. a machinelike) agent. These results are particularly important for companies utilizing AI as direct points of customer contact, with implications for the application and depiction of AI agents.

General Discussion

The present research investigates how consumer responses to offers that are discrepant from expectations depend on whether the administering marketing agent is AI or a human. Our theoretical framework proposes that in the case of worse-than-expected offers, consumers respond more positively when dealing with an AI agent versus a human or more humanlike AI agent, in the form of increased purchase likelihood, satisfaction, and reengagement. In contrast, for better-than-expected offers, consumers respond more positively to a human or more humanlike AI (vs. machinelike AI) agent. Moreover, we propose that this pattern emerges because AI (vs. humans) is inferred to have weaker benevolent and selfish intentions, which in turn influences offer responses.

A series of five studies provides support for our theory across product and service contexts. In each study, an expectation-discrepant offer administered by an AI agent elicits a different response relative to an identical offer administered by a human or more humanlike AI agent. Study 1a demonstrates in a product purchase context that consumers are more likely to accept worse-than-expected offers from an AI agent than a human agent. Study 1b examines the other side of our effect and demonstrates that consumers are more likely to accept better-than-expected offers from a human agent than an AI agent. Study 2 reveals that the effect is driven by stronger inferred intentions on the part of human versus AI agents; that is, stronger benevolent intentions in the case of a better-than-expected offer and stronger selfish intentions in the case of a worse-than-expected offer. These intentions, in turn, alter consumer response. Study 3a reveals that an anthropomorphized (vs. a machinelike) AI agent leads to a lower acceptance of worse-than-expected offers. Finally, Study 3b examines how the physical embodiment of AI (i.e., a robot form) influences anthropomorphism, with implications for responses to discrepant offers. Administration of a better-than-expected offer by a humanlike (vs. machinelike) AI results in higher satisfaction and reengagement intentions. Together, these findings contribute to the literature examining human–AI interactions and technology in marketing, in particular those areas examining preference, service, satisfaction, and anthropomorphism, while identifying important implications for consumers and marketers.

Broad Implications of Differential Responses to AI Versus Human Agents

Our works reveal that fundamental differences exist between the interpretation of negative and positive outcomes administered by AI versus human agents. The underlying mechanisms of perceived intentionality apply to a broad range of interactions beyond just the price discrepancies in our present work. Given this, other marketing and nonmarketing contexts that deal with the administration of negative and positive outcomes will likely demonstrate similar patterns of results as those proposed by H_1–H_3. For example, consumers are commonly faced with unexpected delays or cancellations (e.g., flights, lodging stays, package deliveries), personal evaluations that differ from expectations (e.g., work performance, credit scores), discrepant financial offers (e.g., loan interest rates, credit line amounts), and unexpected school admissions and workplace hiring outcomes, in addition to a broad range of other contexts in which AI plays an increasingly common and customer-facing role. Future research could also examine if the perceived fairness of an offer is affected by the type of agent. For example, the positive or negative response to unexpected offers (e.g., an unexpectedly low or high loan interest rate) may polarize more extremely when administered by a human (vs. AI) (for initial evidence in support of this prediction, see the supplemental analysis of Study 2 in Web Appendix D). Future research could further examine the role of perceived fairness in explaining how perceived intentions influence offer acceptance.

When and How Marketing Managers Should Use Human Agents Versus AI Agents

Two particularly important marketing implications stem directly from the present research, both rooted in the ongoing

transition within firms from human agents to AI agents as marketing representatives. First, our findings give insight into how this transition can be systematically and selectively managed to improve customer purchase tendencies, satisfaction, and reengagement. Indeed, whereas extant works regarding customer-facing AI have revealed primarily main effects of human-to-AI transitions on consumer engagement—that is, either humans or AI are generally superior in a given product or service domain (e.g., Longoni, Bonezzi and Morewedge 2019; Mende et al. 2019)—our work reveals that the nature of the transaction (i.e., offers that are discrepant from expectations) is important. In other words, our findings inform when and where the transition should occur in product and service offer situations involving discrepant expectations. Managers could apply our findings to prioritize (vs. postpone) human-to-AI role transitions in situations where better-than-expected (vs. worse-than-expected) offers are more frequent and impactful. In addition, our results suggest that even when a role transition is not holistically passed to an AI, the selective recruitment of an AI agent to disclose worse-than-expected offers should still be advantageous—that is, our results reveal that an AI "bad cop"/human "good cop" approach to managing discrepant expectations should have beneficial outcomes for the firm. Moreover, our research builds on brand crisis research that explores consumer response to algorithmic errors (Srinivasan and Abi 2021); however, our work reveals differences in perceived intentions, outside the context of algorithmic errors, related to whether an offer is administered by a human or AI.

Second, and perhaps even more impactful, our work has implications for firms that have already transitioned to consumer-facing AI representatives, including the multitude of online and mobile applications that use AI-based algorithms to create and administer offers (e.g., Myer's "6-second sale"; Green 2017). In such posttransition instances where AI is already customer facing, our research reveals that AI may be depicted or otherwise framed as either machinelike or humanlike. Our manipulations show that this can be accomplished through differences in the embodied appearance of the AI (e.g., a robotic form shown in Study 3b). Moreover, the results of Study 3a suggest that the majority of contemporary consumers consider AI to be inherently machinelike (though Study 3a reveals that a minority of consumers do already consider AI to be moderately humanlike) unless humanlike imagery or verbiage is introduced. Thus, marketers should anticipate a default "machinelike" response toward AI agents unless specifically humanlike attributes are depicted. With this understanding, marketers should depict an AI agent as machinelike when disclosing a worse-than-expected offer but should depict more humanlike attributes in the case of a better-than-expected offer.

Ethical Implications

The aforementioned recommendations carry ethical implications, as our findings reveal potential opportunities and concerns for customer–firm relationships and consumer well-being. On the one hand, our work reveals opportunities for firms to receive due, and perhaps otherwise overlooked, credit for better-than-expected offers and potentially other behaviors that are ostensibly benevolent. On the other hand, regarding threats to consumer well-being, our work does reveal a tool through which marketers can increase the acceptance of worse-than-expected offers made to a customer, while also increasing intentions to reengage with the offending firm. In those instances where the worse-than-expected offer is objectively detrimental to consumers, the use of this approach does raise ethical concerns. By documenting these effects, we intend to benefit consumers and policy makers through strengthening their understanding of the nature of differential consumer responses to discrepant outcomes.

Importantly, we propose that this ethical dilemma predominately emerges in situations where consumers are ostensibly harmed—that is, where natural consumer resistance to unexpectedly high prices is bypassed by the selective use of AI. However, some situations do exist where increased consumer purchase would not result in objective consumer harm, and there are even select contexts where consumers and firms could jointly benefit from this effect. For example, consumers often hold unrealistic expectations for product and service offers due to self-serving biases or lack of information. For example, a customer could anchor on a historic promotional price (Lin and Chen 2017) or observe that another customer received a lower price offer without knowledge of that other customer's greater earned loyalty status (Mayser and Wangenheim 2013). Such misperceptions or biased interpretations could lead consumers to irrationally abandon transactions that are objectively beneficial. In such cases, consumers and firms would benefit from shifting consumer tendencies toward preserving the marketing relationship, thereby minimizing switching costs and maximizing consumer utility. Thus, the current research indicates ways managers can strengthen their customer relationships through AI agents when these findings are applied ethically.

The results of our current studies reveal that most consumers currently tend to see AI as not inherently possessed of benevolent or selfish intentions when administering offers. However, there is the possibility that some consumers infer that humans within companies are manipulating AI offers behind the scenes or otherwise have programmed the AI to indirectly carry out selfish human intentions. In such instances, the observed increase in acceptance for worse-than-expected offers from AI agents could be attenuated. Exploring whether certain segments of consumers hold such skeptical beliefs, or whether similar beliefs may increase over time as AI administration of offers becomes even more common, is a fruitful avenue for future research.

Anthropomorphism Theory

Our findings also contribute to the literature on anthropomorphism in product marketing and technology. Our research reveals that anthropomorphism of AI agents—that is, making

AI agents more humanlike—can actually lead to decreased preference and increased disengagement. This is in contrast to previous research, which has shown predominately positive consequences of anthropomorphized products in the form of increased consumer engagement and product liking (Aggarwal and McGill 2007). Whereas extant research suggests that humanlike AI is more comforting and trusted (Longoni, Bonezzi, and Morewedge 2019; Waytz, Heafner, and Epley 2014), our findings reveal situations where humanlike AI may serve as a liability for maintaining the marketing relationship, particularly in contexts of discrepant expectations. Future research should explore other psycholinguistic and personality psychology factors incorporated into AI agent design (e.g., altering the gender or perceived cultural origins of the AI) and resulting changes in consumer perception and reactions.

Beyond contextual effects due to AI design elements, our research also indicates that consumers individually vary in their tendency to anthropomorphize AI agents, with subsequent implications for consumption. Our results reveal that individual differences in perceptions of technology anthropomorphism moderate offer acceptance in cases of worse-than-expected offers (Study 3a). Managers should consider integrating trait measures of technology anthropomorphism when developing consumer marketing segmentations relevant to AI interactions. Moreover, it is possible that other consumer-level personality traits will influence consumer–AI interactions (e.g., big five personality factors)—a potential topic for further research.

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